EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If $a_1, a_2, a_3, ..., a_n, ...$ are in A.P. such that $a_4 - a_7 + a_{10} = m$, then the sum of first 13 terms of this A.P., is:

(A) $15\,m$ (B) $10\,m$ (C) $12\,m$ (D) $13\,m$

MSS001

2. If a, b, c are in AP, then $(a - c)^2$ equals

(A) $4\left(b^2 - ac\right)$ (B) $4\left(b^2 + ac\right)$ (C) $4b^2 - ac$ (D) $b^2 - 4ac$

MSS002

3. Let $a_1, a_2, a_3, ...$ be an A.P. such that $\frac{a_1 + a_2 + ... + a_p}{a_1 + a_2 + a_3 + ... + a_q} = \frac{p^2}{q^2}$; $p \neq q$. Then $\frac{a_6}{a_{21}}$ is equal to:

(A) $\frac{121}{1861}$ (B) $\frac{11}{41}$ (C) $\frac{121}{1681}$ (D) $\frac{41}{11}$

MSS003

4. Let a_n, $n \in N$ is an A.P. with common difference 'd' and all whose terms are non-zero. If n approaches infinity, then the sum $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + ... + \frac{1}{a_n a_{n+1}}$ will approach

(A) $\frac{1}{a_1 d}$ (B) $\frac{2}{a_1 d}$ (C) $\frac{1}{2a_1 d}$ (D) $a_1 d$

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MSS006

5. The first term of an infinite G.P. is 1 and every term is equals to the sum of the successive terms, then its fourth term will be-

(A) $\frac{1}{2}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{16}$

MSS007

6. If $a = \sum_{n=0}^{\infty} x^n, b = \sum_{n=0}^{\infty} y^n, c = \sum_{n=0}^{\infty} (xy)^n$ where $|x|, |y| < 1$; then-

(A) $abc = a + b + c$ (B) $ab + bc = ac + b$
(C) $ac + bc = ab + c$ (D) $ab + ac = bc + a$

MSS008

7. In a GP, first term is 1. If $4T_2 + 5T_3$ is minimum, then its common ratio is

(A) $\frac{2}{5}$ (B) $-\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $-\frac{3}{5}$

MSS009

8. If $\frac{1+3+5+.....\text{upto } n \text{ terms}}{4+7+10+....\text{upto } n \text{ terms}} = \frac{20}{7\log_{10} x}$ and

$n = \log_{10} x + \log_{10} x^{\frac{1}{2}} + \log_{10} x^{\frac{1}{4}} + \log_{10} x^{\frac{1}{8}} + ... + \infty$, then x is equal to

(A) 10^3 (B) 10^5 (C) 10^6 (D) 10^7

MSS011

For More Join: @IITJEE_Advanced

9. If $a \neq 1$ and $\ln a^2 + (\ln a^2)^2 + (\ln a^2)^3 + \ldots\ldots = 3\left(\ln a + (\ln a)^2 + (\ln a)^3 + (\ln a)^4 + \ldots\ldots\right)$ then 'a' is equal to

(A) $e^{1/5}$ (B) $\sqrt{e}$ (C) $\sqrt[3]{e}$ (D) $\sqrt[4]{e}$

MSS012

10. If $x = \sum_{n=0}^{\infty} a^n$, $y = \sum_{n=0}^{\infty} b^n$, $z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in A.P. and $|a| < 1$, $|b| < 1$, $|c| < 1$, then x, y, z are in-

(A) HP (B) Arithmetic - Geometric Progression

(C) AP (D) GP

MSS014

11. If G be the GM between x and y, then the value of $\frac{1}{G^2 - x^2} + \frac{1}{G^2 - y^2}$ is equal to

(A) G^2 (B) $\frac{2}{G^2}$ (C) $\frac{1}{G^2}$ (D) $3G^2$

MSS010

12. If $x \in R$, the numbers $\left(5^{1+x} + 5^{1-x}\right), \frac{a}{2}, \left(25^x + 25^{-x}\right)$ form an A.P. then 'a' must lie in the interval

(A) [1, 5] (B) [2, 5] (C) [5, 12] (D) [12, ∞)

MSS017

13. $2 + 4 + 7 + 11 + 16 + \ldots\ldots\ldots$ to n terms =

(A) $\frac{1}{6}(n^2 + 3n + 8)$ (B) $\frac{n}{6}(n^2 + 3n + 8)$ (C) $\frac{1}{6}(n^2 - 3n + 8)$ (D) $\frac{n}{6}(n^2 - 3n + 8)$

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MSS018

14. The sum $\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \ldots\ldots$ upto 11-terms is :-

(A) $\frac{11}{4}$ (B) $\frac{60}{11}$ (C) $\frac{7}{2}$ (D) $\frac{11}{2}$

MSS019

15. The sum of the series : $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \ldots\ldots$ up to 10 terms, is:

(A) $\frac{22}{13}$ (B) $\frac{18}{11}$ (C) $\frac{20}{11}$ (D) $\frac{16}{9}$

MSS020

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If a_1, a_2 , ..., a_n are distinct terms of an A.P., then

(A) $a_1 + 2a_2 + a_3 = 0$ (B) $a_1 - 2a_2 + a_3 = 0$

(C) $a_1 + 3a_2 - 3a_3 - a_4 = 0$ (D) $a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$

MSS056

2. Let a_1, a_2, a_3 ..... and b_1, b_2, b_3 ..... be arithmetic progressions such that $a_1 = 25$, $b_1 = 75$ and $a_{100} + b_{100} = 100$. Then

(A) the difference between successive terms in progression 'a' is opposite of the difference in progression 'b'.

(B) $a_n + b_n = 100$ for any n.

(C) $(a_1 + b_1), (a_2 + b_2), (a_3 + b_3)$, ... are in A.P.

(D) $\sum_{r=1}^{100}(a_r + b_r) = 10000$

MSS057

3. Real number x_1, x_2, x_3 in A.P. satisfying the equation $x^3 - x^2 + \alpha x + \beta = 0$ then select correct alternative(s)

(A) $\alpha \le \frac{1}{3}$ (B) $\beta \ge -\frac{1}{27}$ (C) $\alpha > \frac{1}{3}$ (D) $\beta < -\frac{1}{27}$

MSS058

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4. If a_1, a_2, a_3 ..... are positive numbers in G.P., then which of the following is/are true -

(A) $a_1^3, a_2^3, a_3^3, \ldots\ldots$ are in G.P. (B) $\frac{a_2}{a_1}, \left(\frac{a_3}{a_2}\right)^2, \left(\frac{a_4}{a_3}\right)^3$ ..... are in G.P.

(C) $\ell n a_1, \ell n a_2, \ell n a_3$..... are in A.P. (D) a_1a_2, a_2a_3, a_3a_4..... are in G.P.

MSS059

5. If equation $16x^4 - mx^3 + (2m + 17)x^2 - mx + 16 = 0$ has four distinct roots forming a geometric progression, and one of the root is $\frac{1}{2}$, then:

(A) common ratio of G.P. is 4 (B) $m = 170$

(C) sum of the roots = $\frac{85}{8}$ (D) Product of root = 3

MSS060

6. Three numbers a, b, c between 2 and 18 are such that

(i) their sum is 25

(ii) the numbers 2, a, b, are in A.P.

(iii) the number b, c, 18 are in G.P.

then which of the following options are correct.

(A) $a = 5$ (B) $b = 8$ (C) $b + c = 20$ (D) $a + c = 17$

MSS061

For More Join: @IITJEE_Advanced

7. If a, b, c are positive integers forming first three terms of increasing geometric progression with common ratio 'r', where $b - a$ is perfect cube and $log_6 a + log_6 b + log_6 c = 6$. Let S denotes sum of 10 terms of another G.P. whose first term is 'a' and common ratio is $\frac{5r}{2}$, then

(A) $a + b + c = 189$ (B) $a + b + c = 273$
(C) sum of digits of S is 90. (D) $S = 10^{10} - 1$

MSS062

8. Let $0 < a < b$ such that $2, 2^x - \frac{7}{2}, 2^x - 5$ are harmonic mean, arithmetic mean and geometric mean of a, b respectively then which of the following option(s) is/are correct

(A) number of possible values of x is 2
(B) number of value of x is 1
(C) $a^2 + b^2$ can be 63
(D) $a^2 + b^2$ can be 3

MSS063

9. The value of $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n}{5^n}$ is $\frac{a}{b}$, where a & b are relatively prime then

(A) $a > b$ (B) $a < b$ (C) $a > 3$ (D) $b > 10$

MSS064

10. The value of $\sum_{r=2}^{100}\frac{2^r(2-r)}{r(r+1)(r+2)}$ is equal to

(A) $\frac{2}{3}-\frac{2^{100}}{(51)(101)}$ (B) $\frac{2}{3}-\frac{2^{101}}{(101)(102)}$ (C) $\frac{1}{2}-\frac{2^{100}}{100.101}$ (D) $\frac{1}{2}-\frac{2^{100}}{(101)(51)}$

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MSS065

11. First three terms of the sequence 1/16, a, b, 1/6 are in geometric series and last three terms are in harmonic series if

(A) $a = \frac{1}{9}, b = \frac{1}{12}$ (B) $a = \frac{1}{12}, b = \frac{1}{9}$ (C) $a = 1, b = -\frac{1}{4}$ (D) $a = -\frac{1}{4}, b = 1$

MSS145

12. For the series $2+\left(\sqrt{2}+\frac{1}{\sqrt{2}}\right)+\left((2\sqrt{2}-1)+\frac{1}{2}\right)+\left((3\sqrt{2}-2)+\frac{1}{2\sqrt{2}}\right)+.....$

(A) $S_n = \sqrt{2}\left(\sqrt{2}+n-1\right) - n + \left(\frac{\left(2^{n/2}-1\right)}{\left(\sqrt{2}-1\right)2^{\frac{n-1}{2}}}\right)$

(B) $T_n = \sqrt{2}\left(\sqrt{2}+n-1\right) - n + \left(\frac{1}{2}\right)^{\frac{n-1}{2}}$

(C) $S_n = \frac{n}{2}\left(3+(n-1)\sqrt{2}-n\right) + \left(\frac{\left(2^{n/2}-1\right)}{\left(\sqrt{2}-1\right)2^{\frac{n-1}{2}}}\right)$

(D) $S_n = \frac{n}{2}\left(3+(n-1)\sqrt{2}-n\right) +$

MSS146

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 13 to 14

A geometric series has first term 'a' and common ratio 'r'. The second term of the series is 4 and the sum upto infinite terms of the series is 25

13. The sum of all possible values of a is-

(A) 5 (B) 15

(C) 20 (D) 25

MSS066

14. If 'r' is taken the larger of its two possible values then the smallest value of n for which S_n exceeds 15 is-

(A) 4 (B) 5

(C) 6 (D) 7

MSS067

Paragraph for Question No. 15 to 16

In a G.P. the sum of the first and last term is 66 and the product of second and the second last term is 128. Sum of all terms is 126

Based on above information, answer the following questions:

15. If an increasing G.P. is considered then number of terms

(A) 2 (B) 4

(C) 6 (D) 8

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MSS070

16. Difference of greatest and least term is

(A) 60 (B) 62

(C) 64 (D) 66

MSS071

Paragraph for Question No. 17 to 18

From first 'n' natural numbers, 2 consecutive numbers are removed the arithmetic mean of the remaining number is $\frac{105}{4}$ then answer the following

17. n is divisible by

(A) 15 (B) 25

(C) 5 (D) 10

MSS068

18. The removed number is(are)

(A) 6 (B) 7

(C) 8 (D) 9

MSS069

For More Join: @IITJEE_Advanced

MATCHING LIST TYPE QUESTION

19. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	Number of common integers in two arithmetic progression 1,3,5,7,...157 and 2,5,8,11...161 is	(1)	25
(Q)	If a and b are roots of $11x^2 - 4x - 2 = 0$, then value of $10(1 + a + a^2 + ...\infty)(1 + b + b^2 + ...\infty)$ is	(2)	26
(R)	If the sum of first 20 terms is same as the sum of first 10 terms in an arithmetic progression then the sum of first 30 terms is less than	(3)	22
(S)	If sum of $\frac{3}{1^2.2^2}+\frac{5}{2^2.3^2}+\frac{7}{3^2.4^2}....+\frac{199}{99^2.100^2}$ is $\frac{a}{b}$ (where a and b are coprime) the value of $20\lvert b - a\rvert$ is	(4)	20
		(5)	10

(A) P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 1,2,3; S $\rightarrow$ 3

(B) P $\rightarrow$ 2; Q $\rightarrow$ 3; R $\rightarrow$ 1,2,3,4,5; S $\rightarrow$ 4

(C) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 1,2; S $\rightarrow$ 2

(D) P $\rightarrow$ 1; Q $\rightarrow$ 4; R $\rightarrow$ 3,4; S $\rightarrow$ 1

MSS072

MATRIX MATCH TYPE QUESTION

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20.

	Column - I		**Column - II**
(A)	The coefficient of x^{49} in the product $(x - 1)(x - 3)(x - 5)(x - 7)$ ..... $(x - 99)$	(p)	–2500
(B)	Let S_n denote sum of first n terms of an A.P. If $S_{2n} = 3S_n$, then $\frac{S_{3n}}{S_n}$ is	(q)	9
(C)	The sum $\sum_{r=2}^{\infty}\frac{1}{r^2-1}$ is equal to	(r)	3/4
(D)	The length, breadth, height of a rectangular box are in G.P. (length > breadth > height) The volume is 27, the total surface area is 78. Then the length is	(s)	6

MSS073

For More Join: @IITJEE_Advanced

EXERCISE (S)

1. The interior angles of a convex polygon form an arithmetic progression with a common difference of 4°. Determine the number of sides of the polygon if its largest interior angle is 172°.

MSS021

2. If the sum of the first 11 terms of an arithmetical progression equals that of the first 19 terms, then the sum of its first 30 terms, is

MSS004

3. The fourth power of the common difference of an arithmetic progression with integer entries is added to the product of any four consecutive terms of it. Prove that the resulting sum is the square of an integer.

MSS147

4. Along a road lies an odd number of stones placed at intervals of 10 m. These stones have to be assembled around the middle stone. A person can carry only one stone at a time. A man carried out the job starting with the stone in the middle, carrying stones in succession, thereby covering a distance of 4.8 km. Then the number of stones is

MSS005

5. The first term of an arithmetic progression is 1 and the sum of the first nine terms equal to 369. The first and the ninth term of a geometric progression with real common ratio coincide with the first and the ninth term of the arithmetic progression. Find the seventh term of the geometric progression.

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MSS022

6. If the pth, q^{th} & r^{th} terms of an AP are in GP. Find the common ratio of the GP.

MSS148

7. Find the 4th term of an H.P. whose 7th term is $\frac{1}{20}$ and 13th term is $\frac{1}{38}$.

MSS149

8. If a, b, c are distinct positive real in H.P., then the value of the expression, $\frac{b+a}{b-a}+\frac{b+c}{b-c}$ is equal to

MSS013

9. Let five geometric means are inserted between $\frac{32}{3}$ and $\frac{243}{2}$ then find sum of all the geometric means.

MSS150

10. Find the sum of the sequence $\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+\ldots\infty$

MSS024

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Let $a_1, a_2, a_3, \ldots\ldots, a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$.

If $a_1^2 + a_2^2 + \ldots\ldots + a_{17}^2 = 140m$, then m is equal to- **[JEE (Main) 2018]**

(1) 68 (2) 34 (3) 33 (4) 66

MSS031

2. Let A be the sum of the first 20 terms and B be the sum of the first 40 terms of the series $1^2 + 2\cdot2^2 + 3^2 + 2\cdot4^2 + 5^2 + 2\cdot6^2 + \ldots\ldots$. If $B - 2A = 100\lambda$, then λ is equal to : **[JEE (Main) 2018]**

(1) 248 (2) 464

(3) 496 (4) 232

MSS032

3. Let $a_1, a_2, \ldots\ldots\ldots, a_{30}$ be an A.P., $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{(2i-1)}$. If $a_5 = 27$ and $S - 2T = 75$, then a_{10} is equal to: **[JEE (Main) 2019]**

(1) 57 (2) 47 (3) 42 (4) 52

MSS033

4. The sum of the following series

$$1 + 6 + \frac{9\left(1^2 + 2^2 + 3^2\right)}{7} + \frac{12\left(1^2 + 2^2 + 3^2 + 4^2\right)}{9} + \frac{15\left(1^2 + 2^2 + \ldots. + 5^2\right)}{11} + \ldots.$$

up to 15 terms, is : **[JEE (Main) 2019]**

(1) 7820 (2) 7830 (3) 7520 (4) 7510

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MSS034

5. Let x, y be positive real numbers and m, n positive integers. The maximum value of the expression $\frac{x^m y^n}{(1+x^{2m})(1+y^{2n})}$ is :- **[JEE (Main) 2019]**

(1) $\frac{1}{2}$ (2) $\frac{1}{4}$ (3) $\frac{m+n}{6mn}$ (4) 1

MSS035

6. If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to : **[JEE (Main) 2019]**

(1) $\beta\gamma$ (2) 0 (3) $\alpha\gamma$ (4) $\alpha\beta$

MSS036

7. If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct? **[JEE (Main) 2019]**

(1) d, e, f are in A.P. (2) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

(3) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P. (4) d, e, f are in G.P.

MSS037

For More Join: @IITJEE_Advanced

8. If $a_1, a_2, a_3, \ldots\ldots, a_n$ are in A.P. and $a_1 + a_4 + a_7 + \ldots\ldots + a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to : **[JEE (Main) 2019]**

(1) 38 (2) 98 (3) 76 (4) 64

MSS038

9. Let a, b and c be in G. P. with common ratio r, where $a \neq 0$ and $0 < r \leq \frac{1}{2}$. If $3a, 7b$ and $15c$ are the first three terms of an A. P., then the 4th term of this A. P. is : **[JEE (Main) 2019]**

(1) $\frac{7}{3}a$ (2) a (3) $\frac{2}{3}a$ (4) $5a$

MSS039

10. If the sum of the first 40 terms of the series, $3 + 4 + 8 + 9 + 13 + 14 + 18 + 19 + \ldots$ is $(102)m$, then m is equal to : **[JEE (Main) 2020]**

(1) 20 (2) 5 (3) 10 (4) 25

MSS040

11. Let $a_1, a_2, a_3, \ldots$ be a G.P. such that $a_1 < 0$, $a_1 + a_2 = 4$ and $a_3 + a_4 = 16$. If $\sum_{i=1}^{9} a_i = 4\lambda$, then λ is equal to : **[JEE (Main) 2020]**

(1) –171 (2) 171 (3) $\frac{511}{3}$ (4) –513

MSS041

12. Five numbers are in A.P., whose sum is 25 and product is 2520. If one of these five numbers is $-\frac{1}{2}$, then the greatest number amongst them is: **[JEE (Main) 2020]**

(1) $\frac{21}{2}$ (2) 27 (3) 16 (4) 7

MSS042

13. The sum, $\sum_{n=1}^{7} \frac{n(n+1)(2n+1)}{4}$ is equal to _____. **[JEE (Main) 2020]**

MSS043

14. The number of terms common to the two A.P.'s 3, 7, 11, ..., 407 and 2, 9, 16, ..., 709 is _____. **[JEE (Main) 2020]**

MSS044

15. The product $2^{\frac{1}{4}} \cdot 4^{\frac{1}{16}} \cdot 8^{\frac{1}{48}} \cdot 16^{\frac{1}{128}} \cdot \ldots$ to ∞ is equal to: **[JEE (Main) 2020]**

(1) $2^{\frac{1}{2}}$ (2) $2^{\frac{1}{4}}$ (3) 2 (4) 1

MSS045

16. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_1 = 1$, $a_2 = 1$ and $a_{n+2} = 2a_{n+1} + a_n$ for all $n \geq 1$. Then the value of $47\sum_{n=1}^{\infty} \frac{a_n}{2^{3n}}$ is equal to _____. **[JEE (Main) 2021]**

MSS046

17. The number of 4-digit numbers which are neither multiple of 7 nor multiple of 3 is ___. **[JEE (Main) 2021]**

MSS048

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For More Join: @IITJEE_Advanced

18. The sum of first four terms of a geometric progression (G.P.) is $\frac{65}{12}$ and the sum of their respective reciprocals is $\frac{65}{18}$. If the product of first three terms of the G.P. is 1, and the third term is α, then 2α is _____. **[JEE (Main) 2021]**

MSS049

19. If for $x, y \in \boldsymbol{R}, x > 0, y = \log_{10} x + \log_{10} x^{1/3} + \log_{10} x^{1/9} + ...$ upto ∞ terms and $\frac{2+4+6+....+2y}{3+6+9+....+3y} = \frac{4}{\log_{10} x}$, then the ordered pair (x, y) is equal to : **[JEE (Main) 2021]**

(1) $(10^6, 6)$ (2) $(10^4, 6)$

(3) $(10^2, 3)$ (4) $(10^6, 9)$

MSS050

20. Let $a_1 = b_1 = 1, a_n = a_{n-1} + 2$ and $b_n = a_n + b_{n-1}$ for every natural number $n \geq 2$. Then $\sum_{n=1}^{15} a_n \cdot b_n$ is equal to _____. **[JEE (Main) 2022]**

MSS051

21. $\frac{2^3-1^3}{1\times7} + \frac{4^3-3^3+2^3-1^3}{2\times11} + \frac{6^3-5^3+4^3-3^3+2^3-1^3}{3\times15} + ... + \frac{30^3-29^3+28^3-27^3+...+2^3-1^3}{15\times63}$ is equal to ____. **[JEE (Main) 2022]**

MSS052

22. If $\frac{1}{2\cdot3^{10}} + \frac{1}{2^2\cdot3^9} + ... \frac{1}{2^{10}\cdot3} = \frac{K}{2^{10}\cdot3^{10}}$, then the remainder when K is divided by 6 is **[JEE (Main) 2022]**

(1) 1 (2) 2 (3) 3 (4) 5

MSS054

23. If $A = \sum_{n=1}^{\infty} \frac{1}{\left(3+(-1)^n\right)^n}$ and $B = \sum_{n=1}^{\infty} \frac{(-1)^n}{\left(3+(-1)^n\right)^n}$, then $\frac{A}{B}$ is equal to: **[JEE (Main) 2022]**

(1) $\frac{11}{9}$ (2) 1 (3) $-\frac{11}{9}$ (4) $-\frac{11}{3}$

MSS055

24. The sum to 10 terms of the series $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + ...$ is:- **[JEE (Main) 2023]**

(1) $\frac{59}{111}$ (2) $\frac{55}{111}$

(3) $\frac{56}{111}$ (4) $\frac{58}{111}$

MSS090

25. If $\frac{1^3+2^3+3^3+.....\text{upto } n \text{ terms}}{1\cdot3+2\cdot5+3\cdot7+.....\text{upto } n \text{terms}} = \frac{9}{5}$, then the value of n is **[JEE (Main) 2023]**

MSS091

For More Join: @IITJEE_Advanced

26. The 4th term of GP is 500 and its common ratio is $\frac{1}{m}$, $m \in N$. Let S_n denote the sum of the first n terms of this GP. If $S_6 > S_5 + 1$ and $S_7 < S_6 + \frac{1}{2}$, then the number of possible values of m is _____

[JEE (Main) 2023]

MSS092

27. Let $a_1, a_2, a_3, \ldots$ be a GP of increasing positive numbers. If the product of fourth and sixth terms is 9 and the sum of fifth and seventh terms is 24, then $a_1a_9 + a_2a_4a_9 + a_5 + a_7$ is equal to ___.

[JEE (Main) 2023]

MSS093

28. If $a_n = \frac{-2}{4n^2 - 16n + 15}$, then $a_1 + a_2 + \ldots + a_{25}$ is equal to : **[JEE (Main) 2023]**

MSS094

(1) $\frac{51}{144}$ (2) $\frac{49}{138}$

(3) $\frac{50}{141}$ (4) $\frac{52}{147}$

29. The sum of the series $\frac{1}{1-3\cdot1^2+1^4} + \frac{2}{1-3\cdot2^2+2^4} + \frac{3}{1-3\cdot3^2+3^4} + \ldots$ up to 10 terms is

[JEE (Main) 2024]

(1) $\frac{45}{109}$ (2) $-\frac{45}{109}$

(3) $\frac{55}{109}$ (4) $-\frac{55}{109}$

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MSS151

30. If three successive terms of a G.P. with common ratio $r(r > 1)$ are the lengths of the sides of a triangle and $[r]$ denotes the greatest integer less than or equal to r, then $3[r] + [-r]$ is equal to :

[JEE (Main) 2024]

MSS152

31. Let ABC be an equilateral triangle. A new triangle is formed by joining the middle points of all sides of the triangle ABC and the same process is repeated infinitely many times. If P is the sum of perimeters and Q is be the sum of areas of all the triangles formed in this process, then:

[JEE (Main) 2024]

(1) $P^2 = 36\sqrt{3}Q$ (2) $P^2 = 6\sqrt{3}Q$

(3) $P^2 = 36\sqrt{3}Q^2$ (4) $P^2 = 72\sqrt{3}Q$

MSS153

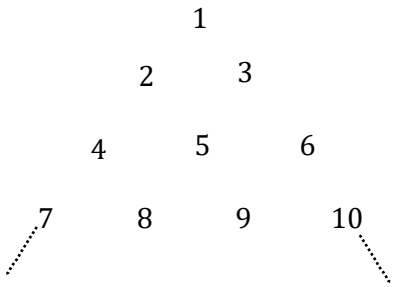
32. Let the first term of a series be $T_1 = 6$ and its rth term $T_r = 3\,T_{r-1} + 6_r$, $r = 2, 3, \ldots, n$. If the sum of the first n terms of this series is $\frac{1}{5}\left(n^2 - 12n + 39\right)\left(4.6^n - 5.3^n + 1\right)$. Then n is equal to _____.

[JEE (Main) 2024]

MSS154

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33. Let the positive integers be written in the form:



If the kth row contains exactly k numbers for every natural number k, then the row in which the number 5310 will be, is _____. **[JEE (Main) 2024]**

MSS155

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EXERCISE - JEE (Advanced) PYQ

1. If the sum of first n terms of an A.P. is cn^2, then the sum of squares of these n terms is **[IIT JEE 2009]**

(A) $\frac{n(4n^2-1)c^2}{6}$ (B) $\frac{n(4n^2+1)c^2}{3}$ (C) $\frac{n(4n^2-1)c^2}{3}$ (D) $\frac{n(4n^2+1)c^2}{6}$

MSS074

2. Let $a_1, a_2, a_3 \ldots\ldots\ldots a_{11}$ be real numbers satisfying
$a_1 = 15$, $27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3,4\ldots\ldots\ldots 11$.
If $\frac{a_1^2 + a_2^2 + \ldots\ldots + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \ldots\ldots + a_{11}}{11}$ is equal to **[IIT JEE 2010]**

MSS075

3. The minimum value of the sum of real numbers a^{-5}, a^{-4}, $3a^{-3}$, 1, a^8 and a^{10} with $a > 0$ is **[IIT JEE 2011]**

MSS076

4. Let $a_1, a_2, a_3, \ldots\ldots\ldots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^{p} a_i, 1 \le p \le 100$. For any integer n with $1 \le n \le 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n, then a_2 is **[IIT JEE 2011]**

MSS077

5. Let $a_1, a_2, a_3, \ldots$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is **[IIT JEE 2012]**

(A) 22 (B) 23 (C) 24 (D) 25

MSS078

6. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s) **[JEE (Advanced) 2013]**

(A) 1056 (B) 1088 (C) 1120 (D) 1332

MSS079

7. A pack contains n cards numbered from 1 to n. Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller to the numbers on the removed cards is k, then $k - 20$ = **[JEE (Advanced) 2013]**

MSS080

8. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $b + 2$, then the value of $\frac{a^2 + a - 14}{a+1}$ is **[JEE (Advanced) 2014]**

MSS081

9. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is 6 : 11 and the seventh term lies in between 130 and 140, then the common difference of this A.P. is **[JEE (Advanced) 2015]**

MSS082

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10. Let $b_i > 1$ for $i = 1, 2, ..., 101$. Suppose $\log_e b_1, \log_e b_2, ..., \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, ..., a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + ... + b_{51}$ and $s = a_1 + a_2 + ... + a_{51}$ then **[JEE (Advanced) 2016]**

(A) $s > t$ and $a_{101} > b_{101}$ (B) $s > t$ and $a_{101} < b_{101}$

(C) $s < t$ and $a_{101} > b_{101}$ (D) $s < t$ and $a_{101} < b_{101}$

MSS083

11. The sides of the right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side? **[JEE (Advanced) 2017]**

MSS084

12. Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11, ... , and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23, ... Then, the number of elements in the set $X \cup Y$ is _____ **[JEE (Advanced) 2018]**

MSS085

13. Let m be the minimum possible value of $\log_3\left(3^{y_1} + 3^{y_2} + 3^{y_3}\right)$, where y_1, y_2, y_3 are real numbers for which $y_1 + y_2 + y_3 = 9$. Let M be the maximum possible value of $(\log_3 x_1 + \log_3 x_2 + \log_3 x_3)$, where x_1, x_2, x_3 are positive real numbers for which $x_1 + x_2 + x_3 = 9$. Then the value of $\log_2(m^3) + \log_3(M^2)$ is _____. **[JEE (Advanced) 2020]**

MSS086

14. Let $a_1, a_2, a_3, ...$ be a sequence of positive integers in arithmetic progression with common difference 2. Also, let $b_1, b_2, b_3, ...$ be a sequence of positive integers in geometric progression with common ratio 2. If $a_1 = b_1 = c$, then the number of all possible values of c, for which the equality $2(a_1 + a_2 + \cdots + a_n) = b_1 + b_2 + ... + b_n$ holds for some positive integer n, is _____

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[JEE (Advanced) 2020]

MSS087

15. Let $\ell_1, \ell_2, ..., \ell_{100}$ be consecutive terms of an arithmetic progression with common difference d_1, and let $w_1, w_2, ..., w_{100}$ be consecutive terms of another arithmetic progression with common difference d_2, where $d_1 d_2 = 10$. For each $i = 1, 2, ..., 100$, let R_i be a rectangle with length ℓ_i, width w_i and area A_i. If $A_{51} - A_{50} = 1000$, then the value of $A_{100} - A_{90}$ is. **[JEE (Advanced) 2022]**

MSS088

16. Let $75\overset{r}{...}57$ denote the $(r+2)$ digit number where the first and the last digits are 7 and the remaining r digits are 5. Consider the sum $S = 77 + 757 + 7557 + ... + 75\overset{98}{...}57$. If $S = \dfrac{75\overset{99}{...}57 + m}{n}$, where m and n are natural numbers less than 3000, then the value of $m + n$ is

[JEE (Advanced) 2023]

MSS089

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	B	A	B	C	B	B	D	A
Que.	11	12	13	14	15					
Ans.	C	D	B	D	C					

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,D	A,B,C,D	A,B	A,B,C,D	A,B,C	A,B,C,D	A,C,D	B,C	B,C,D	A,B
Que.	11	12	13	14	15	16	17	18	19	
Ans.	B,D	B,C	D	B	C	B	B,C,D	B,C	B	
Que.	20									
Ans.	A → p ; B → s ; C → r; D → q									

EXERCISE (S)

1. 12 **2.** 0 **4.** 31 **5.** 27 **6.** $\frac{q-r}{p-q}$

7. $\frac{1}{11}$ **8.** 2 **9.** 211 **10.** 0.50

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	4	1	2	1	3	3	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	504	14	1	7	5143	3	4	27560
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	120	4	3	2	5	12	60	3	4	1
Que.	31	32	33							
Ans.	1	6	103							

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	0	8	9 or 3	D	A,D	5	4	9	B
Que.	11	12	13	14	15	16				
Ans.	6	3748	8.00	1.00	18900	1219				

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Important Notes

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